

Mini Project: Car Price Prediction Using Linear Regression

Goal: Predict the selling price of a car using features like year, kilometers driven, fuel type, etc.

Dataset: Available on Kaggle - 'car data.csv' (CarDekho dataset)

Step 1: Add Title and Description (Markdown)

Explain what the dataset is and what you're predicting - car selling price.

Step 2: Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

Step 3: Load the Dataset

```
df = pd.read_csv('car data.csv')
```

```
df.head()
```

Step 4: Explore the Dataset

```
df.info()
```

```
df.describe()
```

```
df.isnull().sum()
```

```
sns.heatmap(df.corr(), annot=True)
```

Step 5: Preprocess the Data

```
# Drop 'Car_Name' column
```

```
df = df.drop('Car_Name', axis=1)
```

```
# Convert categorical variables to dummy/one-hot encoding
```

```
df = pd.get_dummies(df, drop_first=True)
```

Step 6: Split into Features and Target

```
X = df.drop('Selling_Price', axis=1)
```

```
y = df['Selling_Price']
```

Step 7: Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 8: Train a Linear Regression Model

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

Step 9: Predict and Evaluate

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
print("R2 Score:", r2_score(y_test, y_pred))
```

```
print("MSE:", mean_squared_error(y_test, y_pred))
```

Step 10: Wrap Up (Markdown)

Write a summary of what you did, your model's performance, and key takeaways.